

Answer all questions

- 1 (a) A sample of bromine was found to contain 50.52 % of ^{79}Br and the remainder ^{81}Br .

(i) Name the species ^{79}Br and ^{81}Br .

(ii) Define the term *relative atomic mass*.

(iii) Calculate the relative atomic mass of Bromine.

[4]

- (b) (i) Write the equation that represents the second ionization energy, of oxygen.

(ii) State and explain any two factors that affect ionization energy of an element.

factor _____

explanation _____

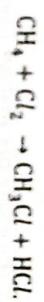
factor _____

explanation _____

[6]
[Total:10]

- 2 (a) (i) Define the term *bond energy*.

- (ii) Methane reacts with chlorine according to the equation



Calculate the enthalpy change for the reaction.

[4]

- (b) (i) State **two** factors that affect the magnitude of lattice enthalpy.

1. _____

2. _____

- (ii) Explain how the factors in (b)(i) affect the magnitude of lattice enthalpy.

[4]

- (c) Explain how lattice enthalpy affects solubility of ionic compounds.

[2]

[Total:10]



3 (a) (i) Write the equation for the decomposition of magnesium carbonate.

(ii) State the use of any one of the products from the reaction in (i).

(iii) Describe and explain the trend in the decomposition of Group (II) carbonates.

[3]

(b) Write chemical equations to show how sulphuric acid reacts with

(i) $\text{KBr}_{(s)}$

[2]

(ii) $\text{NaCl}_{(s)}$

[1]



(c) State the observations made in the reaction in

(i) b(i). _____

(ii) b(ii). _____

[2]
[Total: 10]

4 (a) Outline how propylamine can be prepared from chloroethane.

[3]

(b) Compare the basicity of propylamine with that of phenylamine.

[2]

(c) (i) Write an equation to show how propylamine dissolves in water.

(ii) State the acid/base nature of the solution formed in (i).

[2]

6

(a)

Fig.6.1 shows the apparatus used to separate a mixture of ortho-nitrophenol and para-nitrophenol.

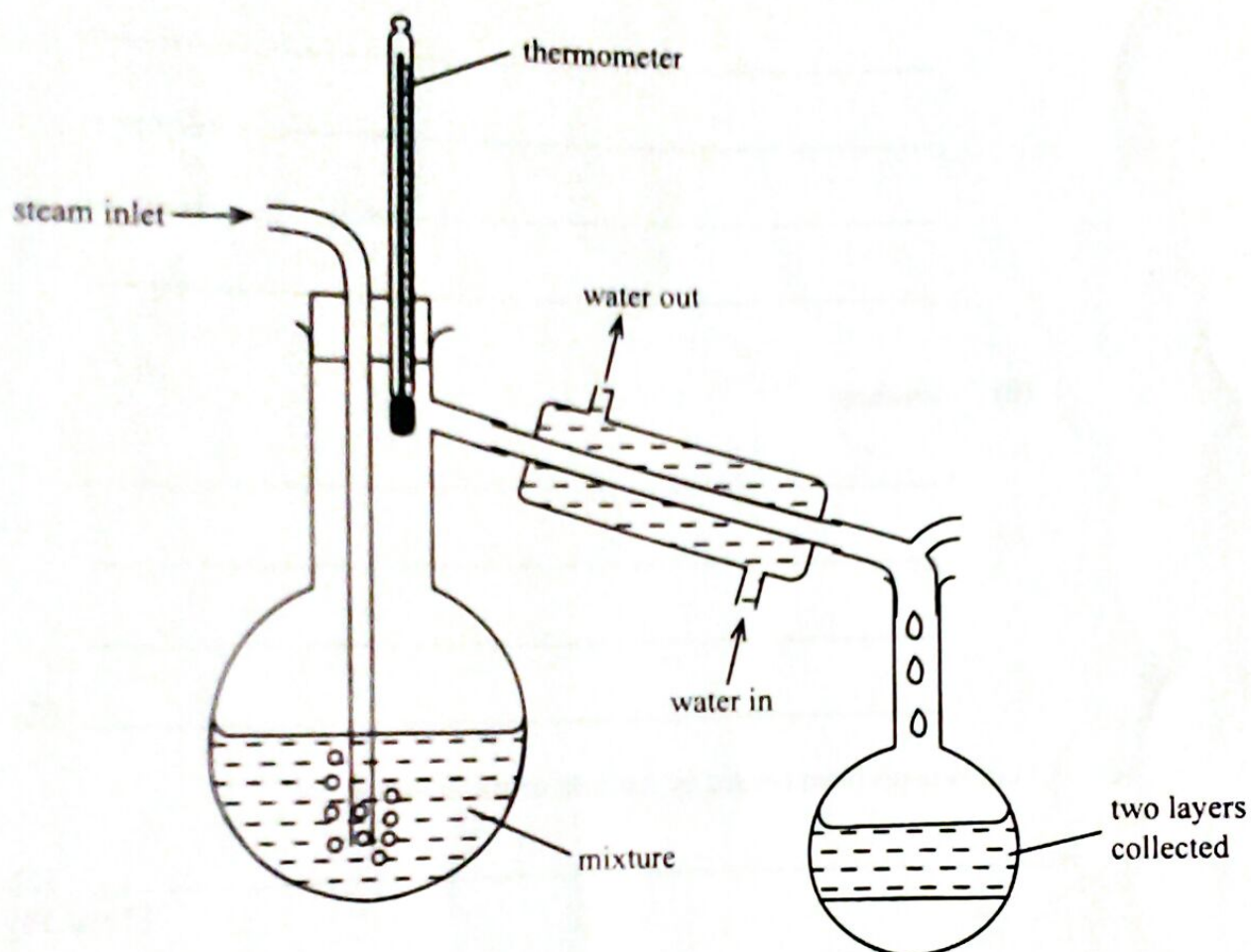


Fig.6.1

(i) Name the separating technique shown.

(ii) State, with reasons, the composition of the distillate.

[6]

(b) Explain the formation in landfills of leachate and methane.

(i) *leachate*

(ii) *methane*

[3]

(c) State **one** problem caused by leachate on the environment.

[1]

[Total:10]

- (d) Superglue contains an organic compound A, shown in Fig.4.1.

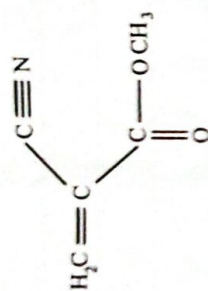


Fig.4.1

- (i) Name any **one** functional group in compound A.
-
- (ii) State the type of polymerization that compound A can undergo.
-
- (iii) Draw the repeat unit of the polymer of A.
-

[3]
[Total: 10]

- 5 (a) The structural formula of Jasmine is shown in Fig.5.1

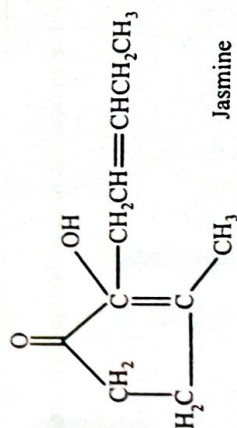


Fig.5.1

Draw the structural formula of the organic product formed when Jasmine reacts with

(i) 2,4-dinitrophenylhydrazine,

(ii) LiAlH_4 ,

(iii) cold dilute KMnO_4 ,

(iv) hot, concentrated KMnO_4 .

[4]